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LOOPBACK DUST CAP FOR FIBER OPTIC CABLE ASSEMBLY AND METHOD OF MAKING SAME

Abstract

A loopback dust cap for a fiber optic cable assembly includes a body and an inserted received in a cavity of the body. The insert supports first and second sleeves, which in turn support first and second cap ferrules that terminate opposite ends of a loopback optical fiber that extends within the body to form a loopback. The first and second sleeves are configured to align ferrules of a fiber optic connector with the first and second cap ferrules and thereby allow the loopback optical fiber to receive and return optical signals to the fiber optic connector.

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Background/Summary

PRIORITY APPLICATION [0001] This application claims the benefit of priority of U.S. Provisional Application No. 63/560,981, filed on Mar. 4, 2024, the content of which is relied upon and incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to optical connectivity, and more particularly to a method and device for testing fiber optic cables.

BACKGROUND

[0003] Optical fibers are useful in a wide variety of applications, including the telecommunications industry for voice, video, and data transmissions. Benefits of optical fibers include wide bandwidth and low noise operation. Continued growth of the Internet has generated demand for data centers having larger numbers of computer systems and associated telecommunication and storage systems. These systems are often connected by one or more fiber optic cables that were installed well before the systems. If a fiber optic cable has excessive signal loss (e.g., due to unlatched or dirty connectors, a severely bent fiber, or any other reason) the systems connected by that fiber optic cable may not function properly. When a technician becomes aware of a problematic optical link, they must identify the fiber optic cable providing the link, track down each end of the fiber optic cable, and then test the fiber optic cable, e.g., using a power meter or installed hardware (e.g., switch) to determine if there is excessive signal loss.

[0004] In large data centers having thousands of fiber optic cables, finding and testing a fiber optic cable providing a defective link between network nodes can be time consuming. Thus, there is a need in the fiber optic industry for improved methods and devices for evaluating fiber optic cables prior to the cables being used to connect network nodes.

SUMMARY

[0005] The present disclosure provides a loopback dust cap and methods of using the loopback dust cap. The loopback dust cap is intended for a fiber optic cable assembly that includes a first optical fiber, a second optical fiber, and a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber. According to one embodiment, the loopback dust cap includes a body that has a front end, a back end, and a cavity between the front end and the back end. The back end of the body includes an opening into the cavity so that the cavity is configured to receive the fiber optic connector. The loopback dust cap also includes an insert received in the cavity of the body and first and second sleeves that are supported by the insert. Each of the first and second sleeves includes a respective first end facing the front end of the body and a respective second end facing the back end of the body. A first cap ferrule extends through the first end of the first sleeve and includes a first end face positioned within the first sleeve. A second cap ferrule extends through the first end of the second sleeve and includes a second end face positioned within the second sleeve. The loopback dust cap also includes a loopback optical fiber having a first end terminated by the first cap ferrule and a second end terminated by the cap second ferrule. The loopback optical fiber extends between the first cap ferrule and the second cap ferrule in a region of the body that is between the front end of the body and the insert. The first sleeve and the second sleeve are configured to: (i) receive the first connector ferrule and the second connector ferrule, respectively, when the fiber optic connector is received in the cavity; and (ii) align the first connector ferrule and the second connector ferrule with the first cap ferrule and the second cap ferrule, respectively.

[0006] The body and insert of the loopback dust cap may be configured to define a mating interface similar to a fiber optic adapter. In some embodiments, for example, the body and the insert may be shaped to define an LC-type interface according to IEC 61754-20:2012 or TIA-604-10-C.

Regardless of whether an intermatability interface is defined, in some embodiments the first cap ferrule and the second cap ferrule may each comprise a ceramic ferrule having a nominal diameter of 1.25 mm.

[0007] In some embodiments, the loopback dust cap further includes: a first ferrule holder received over a portion of the first cap ferrule that is not within the first sleeve; and a second ferrule holder received over a portion of the second cap ferrule that is not within the second sleeve. The first ferrule holder and the second ferrule holder are at least partially retained within the insert. One possible way to achieve this is for the insert to include a support wall, a first boss extending from the support wall toward the front end of the body, and a second boss extending from the support wall toward the front end of the body. The first boss extends over and beyond the first end of the first sleeve, and receives and retains at least a portion of the first ferrule holder. The second boss extends over and beyond the first end of the second sleeve, and receives and retains at least a portion of the second ferrule holder. Optionally, the first boss and the second boss may each comprise respective split end portions configured to expand radially to allow insertion of the first ferrule holder and the second ferrule holder, respectively, when the insert is not received in the cavity. In such embodiments the body may be configured to limit or prevent the split end portions from expanding radially when the insert is received in the cavity.

[0008] In some embodiments, the loopback optical fiber may meet bend performance specifications of International Telecommunication Union standards ITU-T G.657.A2 and/or ITU-T G.657.B2. In other embodiments, the loopback optical fiber may meet bend performance specifications of International Telecommunication Union standard ITU-T G.657.B3.

[0009] According to another aspect of this disclosure, the body of the loopback dust cap may comprise a translucent or transparent material, and the loopback optical fiber may be configured to emit visible light.

[0010] A fiber optic cable assembly including the loopback dust cap is also provided in this disclosure. The fiber optic cable assembly comprises: a first optical fiber and a second optical fiber; a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber; and the loopback dust cap (e.g., such as described in the above paragraphs of this Summary section). The first connector ferrule extends into the first sleeve of the loopback dust cap and contacts the end face of the first cap ferrule, and the second connector ferrule extends into the second sleeve of the loopback dust cap and contacts the end face of the second cap ferrule.

[0011] A method of making a loopback dust cap is also provided in this disclosure, with the loopback dust cap being for a fiber optic cable assembly that includes a first optical fiber, a second optical fiber, and a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber. The method comprises: (a) assembling first and second sleeves with an insert so that the first and second sleeves are supported by the insert; (b) providing a loopback assembly that includes a loopback optical fiber having one end terminated by a first cap ferrule and another end terminated by a second cap ferrule; (c) extending the first cap ferrule through a first end of the first sleeve so that an end face of the first cap ferrule is positioned within the first sleeve; (d) extending the second cap ferrule through a first end of the second sleeve so that an end face of the second cap ferrule is positioned within the second sleeve; and (e) positioning the insert in a cavity of a body. The body includes a front end and a back end, and the back end includes an opening into the cavity. The positioning step (e) occurs after steps (a)-(d) so that the positioning results in: (i) the respective first ends of the first and second sleeves facing the front end of the body and respective second ends of the first and second sleeves facing the back end of the body, and (ii) the loopback optical fiber extending between the first cap ferrule and the second cap ferrule in a region of the body that is between the front end of the body and the insert. The cavity of the body is configured to receive the fiber optic connector through the opening in the back end of the body. Additionally, the first sleeve and the

second sleeve are configured to: (i) receive the first connector ferrule and the second connector ferrule, respectively, when the fiber optic connector is received in the cavity, and (ii) align the first connector ferrule and the second connector ferrule with the first cap ferrule and the second cap ferrule, respectively.

[0012] Additional features and advantages will be set out in the detailed description which follows, and in part will be readily apparent to those skilled in the technical field of optical connectivity. It is to be understood that the foregoing general description, the following detailed description, and the accompanying drawings are merely exemplary and intended to provide an overview or framework to understand the nature and character of the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The accompanying drawings are included to provide a further understanding and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiment(s), and together with the description serve to explain principles and operation of the various embodiments. Features and attributes associated with any of the embodiments shown or described may be applied to other embodiments shown, described, or appreciated based on this disclosure.

[0014] FIG. 1 is perspective view of an exemplary fiber optic connector with a loopback dust cap according to one embodiment of this disclosure.

[0015] FIG. 2 is an exploded perspective view of the exemplary fiber optic connector and loopback dust cap of FIG. 1, with the loopback dust cap itself being exploded into a cap and an insert sub-assembly.

[0016] FIG. 3 is an exploded perspective view of the insert sub-assembly of FIG. 2.

[0017] FIG. 4 is a cross-sectional view of a portion of the insert sub-assembly of FIG. 2.

[0018] FIG. 5 is a different cross-sectional view of the insert sub-assembly of FIG. 2.

[0019] FIGS. 6 and 7 are perspective views showing how the insert sub-assembly of FIG. 2 can be received in a cavity of the body of the loopback dust cap.

[0020] FIG. 8 is a perspective view of an underside of the insert sub-assembly of FIG. 2 prior to positioning the insert sub-assembly in the body.

[0021] FIG. 9 is a cross-sectional view of the loopback dust cap of FIG. 1 installed on the fiber optic connector, wherein only a portion of the fiber optic connector is shown.

DETAILED DESCRIPTION

[0022] Various embodiments will be further clarified by examples in the description below. In general, the description relates to a loopback dust cap for a fiber optic cable assembly that provides an optical loopback which can be used for testing the fiber optic cable assembly and/or an optical link that includes the fiber optic cable assembly. For example, the loopback dust cap may be configured to receive an optical signal (e.g., a 1300 nm optical signal) transmitted down one optical fiber of a fiber optic cable assembly to which the loopback dust cap is attached, and redirect the received optical signal down another optical fiber of the fiber optic cable assembly. The loopback dust cap may thereby provide a loopback to the device (e.g., transceiver) transmitting the optical signal at an opposite end of the fiber optic cable assembly or opposite end of an optical link that includes the fiber optic cable assembly.

[0023] The amount of attenuation in the optical loopback can be determined by comparing the power of the return optical signal to the power of the transmitted optical signal. Attenuation issues in an optical link that includes the fiber optic cable assembly, such as issues that may be caused by the presence of debris or loose connections, can then be identified based on the amount of attenuation. In one embodiment, the loopback dust cap may have a simplified construction,

comprising just two primary molded components—a body and an insert—that can be assembled together, with insert configured to hold a loopback assembly that fits within the body.

[0024] Loopback dust caps and methods of using loopback dust caps to evaluate optical links may be particularly useful in data center systems. Data centers often include duplex optical links in which optical signals are transmitted by one optical fiber and received by another optical fiber. In this environment, loopback dust caps can be placed at one end of a duplex link so that the transmitted signal is returned as the received signal, which can then be measured at the other end, e.g., at a transceiver. Data center optical links often include multiple jumpers, modules, trunk cables, and harnesses. When a loopback dust cap is installed at the end of an optical link, the loopback feature may facilitate determining the loss through that optical link. By comparing losses between optical links, a technician may identify an optical link having a higher-than-normal loss as having a problem. For example, if a relatively large number of optical links (e.g., 99 links) have a relatively low loss (e.g., a loss of less than 1.5 dB), and only a few optical links (e.g., one link) have a relatively high loss (e.g., a loss in excess of 4 dB), the optical links with the outlying attenuations likely have problems.

[0025] FIGS. 1 and 2 illustrate an example embodiment of a fiber optic cable assembly **10** together with a loopback dust cap **20** according to this disclosure. More specifically, FIG. 1 illustrates an end portion of the fiber optic cable assembly **10** that includes a fiber optic connector **12** (“connector **12**”) terminating a fiber optic cable **14**. The loopback dust cap **20** is received on the end of the connector **12**. In FIG. 2, the loopback dust cap **20** is shown removed from the connector **12** and in a partially exploded form. The loopback dust cap **20** includes the two primary molded components mentioned above: a body **22** (or “shell **22**”) and an insert **24**. The insert **24** is part of an insert sub-assembly **26**, which will be described in greater detail below.

[0026] The connector **12** is shown in the form of a form of a duplex LC connector (e.g., according to IEC standard 61754-20:2012 and/or TIA 607-10-C). Accordingly, the loopback dust cap **20** can be considered a duplex LC dust cap since it is configured to be received on the connector **12**. However, the present disclosure is not limited to such types of connectors and dust caps, as persons skilled in optical connectivity will appreciate that many other embodiments are possible. For example, the principles described below for the loopback dust cap **20** may apply to dust cap designs for other duplex connectors, including very small form factor (VSFF) duplex connectors such as SN connectors commercially available from Senko Advanced Components, Inc. and MDC connectors commercially available from US Conec Ltd.

[0027] The connector **12** includes two connector sub-assemblies **30a**, **30b** that each have a ferrule **32** (also referred to herein as “connector ferrule **32**”) extending from a connector body **34**. The connector ferrules **32** terminate different optical fibers that are carried by the fiber optic cable **14**, as is well known. The connector bodies **34** each include a latch arm **36** for engaging an adapter or receptacle (not shown), as is also well-known. The other components of the connector **12** in FIGS. 1 and 2, including a shell **38**, a common housing (hidden from view in FIGS. 1 and 2) covered by the shell **38**, trigger **40** for actuating the latch arms **36**, and boot **42**, are not pertinent to this disclosure and will not be described. Again, the connector **12** is merely an example of a connector for a loopback dust cap according to this disclosure. The loopback dust cap **20** may be used with other designs of duplex LC connectors, and embodiments are possible for other types of connectors such as SN and MDC connectors.

[0028] The connector **12** generally extends along a longitudinal axis A centered between the connector sub-assemblies **30a**, **30b**. In this disclosure, references to an “axial direction”, “axially”, or the like refer to along or parallel to the longitudinal axis A. Furthermore, the terms “front” and “back” (or “rear”) are relative terms that generally use the orientation of the connector **12** along the longitudinal axis A as a reference. For example, a front of the connector **12** is defined by the connector sub-assemblies **30a**, **30b** (and specifically the connector ferrules **32**), and a rear of the connector **12** is defined where the boot **42** of the connector **12** stops extending over the cable **14**.

The loopback dust cap **20** is described below with the same orientation in mind. In other words, a front of the loopback dust cap **20** faces the same direction as the front of the connector **12**, and a back of the loopback dust cap **20** faces the same direction as the rear of the connector **12**.

[0029] The loopback dust cap **20** includes the body **22** and insert sub-assembly **26**, as mentioned above. FIG. 3 illustrates the components of the insert sub-assembly **26** in further detail. As shown in FIG. 3, the insert sub-assembly **26** includes the insert **24**, first and second sleeves **46a**, **46b**, and a loopback assembly **50**. The insert **24** may be a molded component with a main support wall **52** and bosses **54**, **56** extending from opposite sides of the support wall **52**. More specifically, first and second bosses **54a**, **54b** extend from a front side of the support wall **52**, and another pair of bosses **56a**, **56b** (only one boss **56b** being visible in FIG. 3) extend from a back side of the support wall **52**. The first boss **54a** is aligned with one of the bosses **56a** on the back side of the support wall **52**, and the second boss **54b** is aligned with the other boss **56b** on the back side of the support wall **52**. This allows each aligned pair of bosses **54**, **56** to define different portions of a continuous passage **58** (FIG. 4) that also extends through the support wall **52**. The bosses **54**, **56** are generally cylindrical in the embodiment shown and help guide ferrules (including the connector ferrules **32**) into the first and second sleeves **46a**, **46b**, as will be described in greater detail below. To this end, the insert **24** may be similar to inserts used in fiber optic adapters (or to internal portions of a one-piece, molded adapter body). The insert **24** may also include a divider wall **60** that extends from the back side of the support wall **52** between the bosses **56a**, **56b**.

[0030] The first and second sleeves **46a**, **46b** may be slit sleeves like those commonly used in fiber optic adapters for aligning the ferrules of two mated fiber optic connectors. Although shown exploded from the insert **24** in FIG. 3, the first and second sleeves **46a**, **46b** are normally received in the passages **58** (FIG. 4) of the insert **24** and supported by the bosses **56a**, **56b**. In other embodiments, the bosses **54** may be configured to support first and second sleeves **46a**, **46b** instead of or in addition to the bosses **56a**, **56b**.

[0031] The loopback assembly **50** includes first and second cap ferrules **64a**, **64b** that terminate opposite ends of a loopback optical fiber **66**. In the embodiment shown, the first and second cap ferrules **64a**, **64b** are ceramic ferrules that have a nominal diameter of 1.25 mm, similar to the connector ferrules **32** (as is known for LC-type connectors and many other types of connectors that use similar ferrules). The first and second cap ferrules **64a**, **64b** are configured to be received in the first and second sleeves **46a**, **46b**, respectively, and each include an end face **68** that presents an end of the loopback optical fiber **66** for optical coupling. The first and second cap ferrules **64a**, **64b** extend along respective first and second optical axes when the loopback dust cap **20** is assembled. The loopback assembly **50** also includes first and second ferrule holders **70a**, **70b** from which the first and second cap ferrules **64a**, **64b** extend. Thus, the ferrule holders **70a**, **70b** receive and support end portions of the first and second cap ferrules **64a**, **64b**. Each ferrule holder **70a**, **70b** includes a flange portion **72** that defines an axially-facing seating/flange surface **74**.

[0032] FIG. 4 is a cross-sectional view of a portion of the insert sub-assembly **26** in an assembled state, and shows the relationship between the first cap ferrule **64a**, first ferrule holder **70a**, first sleeve **46a**, and bosses **54a**, **56a**. Although not shown in FIG. 4, the second cap ferrule **64b**, second ferrule holder **70b**, second sleeve **46b**, and bosses **54b**, **56b** are arranged in a similar manner. Thus, the following description of FIG. 4 applies equally to the second cap ferrule **64b**, second ferrule holder **70b**, second sleeve **46b**, and bosses **54b**, **56b**.

[0033] As shown in FIG. 4, the support wall **52** and boss **54a** support a portion of the first sleeve **46a**, with a remainder of the first sleeve **46a** extending within the first boss **54a**. The first sleeve **46a** may, for example, be inserted into the passage **58** through an end of the first boss **54a** (which defines an opening **78** into the passage **58**) and advanced through the support wall **52** and into the boss **56a** during an assembly process. The boss **56a** includes an internal shoulder **80** (or “stop **80**”) that the first sleeve **46a** abuts to position the first sleeve **46a** within the insert **24**.

[0034] The portion of the first sleeve **46a** that extends within the first boss **54a** receives and

supports the first cap ferrule **64a**, as shown in FIG. **4**. The first ferrule holder **70a** is also at least partially positioned within the passage **58**. Although the opening **78** into the passage **58** is smaller than the flange portion **72** of the first ferrule holder **70a**, the first boss **54a** includes a split end portion **82** (FIG. **3**) to allow the flange portion **72** to be inserted through the opening **78** and into the passage **58**. The split end portion **82** effectively is formed by slots **84** (two in the embodiment shown) that extend from the end of the first boss **54a** toward the support wall **52**. The slots **84** split the end portion **82** of the first boss **54a** into different arcuate segments so that the first boss **54a** can expand radially to allow the first ferrule holder **70a** to be inserted into the passage **58** until the flange portion **72** reaches a larger-diameter section of the passage **58**, at which point the first boss **54a** can return to its non-expanded configuration. The change in diameter of the passage **58** defines an internal shoulder **86** (or “stop **86**”) in the first boss **54a**.

[0035] As shown in FIG. **4**, in an assembled configuration, the seating surface **74** of the first ferrule holder **70a** faces the internal shoulder **86** of the first boss **54a**. When the seating surface **74** contacts the internal shoulder **86**, the end face **68** of the first cap ferrule **64a** is positioned at or substantially at an optical reference plane OP that is transverse to the axial direction. The optical reference plane OP will be discussed in further detail below in connection with the loopback dust cap **20** being received on the connector **12**. Persons skilled in optical connectivity, however, will appreciate how the first ferrule holder **70a** has a fixed relationship with the first cap ferrule **64a**, and that the seating surface **74** serves as a mechanical reference or datum relative to the end face **68** of the first cap ferrule **64a**. The first boss **54a** can then be designed so that the internal shoulder **86** positions the end face **68** of the first cap ferrule **64a** at or substantially at a desired plane (the optical reference plane OP) when the seating surface **74** of the first ferrule holder **70a** contacts the internal shoulder **86**.

[0036] The cross-section of the first ferrule holder **74a** in FIG. **4** shows the seating surface **74** on the bottom of the flange portion **72** but not the top. This is due to the plane of the cross-section extending through a notch or keyway **90** (FIG. **3**) that is provided on the top of the first ferrule holder **70a**. As shown in FIG. **5**, the first boss **54a** includes a keying feature **92** that is received in the keyway **90** to limit the first ferrule holder **70a** from rotating relative to the first boss **54a** about the first optical axis along which the first ferrule **64a** extends. In alternative embodiments, however, the first ferrule holder **70a** and first boss **54a** may not include complementary keying features. This may be the case if the end face **66** of the first cap ferrule **64a** has ultra physical contact (UPC) polishing geometry, for example.

[0037] FIGS. **6** and **7** illustrate the insert sub-assembly **26** being installed in the body **22** of the loopback dust cap **20**. The body includes a front end **100**, a back end **102**, and a cavity **104** between the front end **100** and the back end **102**. As shown in FIG. **6**, the back end **102** includes an opening **106** into the cavity **104**. The opening **106** may have a shape/profile that generally corresponds to the shape/profile of the support wall **52** of the insert **24**, and the cavity **104** may maintain that shape/profile for at least some length as the cavity **104** extends into the body **22** from the opening **106**. In the embodiment shown, the cavity **104** includes an internal ridge or shoulder **110** at a location spaced from the back end **102**. The internal shoulder **110** serves as a stop to position the insert sub-assembly **26** in the cavity **104**. More specifically, the insert sub-assembly **26** may be passed through the opening **106** and into the cavity **104** until the support wall **52** of the insert **24** contacts the internal shoulder **110**, which is configured to prevent further insertion. This results in the arrangement shown in FIG. **7**, with the insert sub-assembly **26** properly positioned in the body **22** so that the loopback dust cap **20** can be used with the connector **12**.

[0038] Assembling the components in the manner described above may result in the insert sub-assembly **26** being coupled to the body **22**. For example, in the embodiment shown, the body **22** includes a lower channel **114** extending from the back end **102** and a pocket or recess **116** at the end of the lower channel **114**. The lower channel **114** accommodates a catch **118** (FIG. **8**) that is on a bottom side of the insert **24** when the insert **24** is being inserted into the cavity **104** of the body

22. By the time the insert **24** contacts the internal shoulder **110** of the body **22**, the catch **118** is received in the recess **116**. The components are designed such that the catch **118** and recess **116** provide a snap-fit coupling between the insert **24** and the body **22**. Other ways of implementing a snap-fit are possible, however, as are other forms of coupling in general. For example, in alternative embodiments the insert **24** may be coupled to the body **22** using adhesive, ultrasonic welding, and/or another technique.

[0039] Regardless of the coupling technique, when the loopback dust cap **20** is assembled, the body **22** and the insert **24** collectively define an interface for receiving the connector **12** (FIGS. **1** and **2**). The interface may correspond to an intermatability interface for the type of connector for which the loopback dust cap **20** is intended. For the embodiment shown, for example, the loopback dust cap **20** may define an LC-type interface according to intermatability standards IEC 61754-20:2012 or TIA-604-10-C. The optical reference plane OP mentioned above in connection with FIG. **3** may therefore correspond to the optical reference plane associated with the intermatability standards IEC 61754-20:2012 or TIA-604-10-C. Cutouts or slots **122** (FIGS. **1** and **2**) are also provided in the body **22** of the loopback dust cap **20** at a location that corresponds to a mechanical reference plane associated with the intermatability standards, and thereby define latching features for engaging corresponding latching features **124** on the latch arms **36** of the connector sub-assemblies **30a**, **30b**.

[0040] FIG. **9** is a cross-sectional view of the loopback dust cap **20** received on the end of the connector **12**. As shown in the figure, the connector ferrules **32** are respectively received in the first and second sleeves **46a**, **46b** and contact the end faces **68** of the first and second cap ferrules **64a**, **64b**. More specifically, when placing the loopback dust cap **20** on the connector **12**, the connector ferrules **32** enter the respective first and second sleeves **46a**, **46b** and eventually contact the end faces **66** of the first and second cap ferrules **64a**, **64b**. The first and second cap ferrules **64a**, **64b** may be pushed axially by the connector ferrules **32** until the flange portions **72** of the first and second ferrule holders **70a**, **70b** contact respective internal shoulders **86**, which then effectively sets the positions of the first and second cap ferrules **64a**, **64b** within the loopback dust cap **20** in the manner described above with reference to FIG. **3**. The loopback dust cap **20** may still need to be pushed further onto the connector **12** to ensure that the latching features **124** (FIG. **1**) of the latch arms **36** engage the corresponding latching features (e.g., slots **122**) on the body **22** of the loopback dust cap **20**. Because the first and second cap ferrules **64a**, **64b** prevent the connector ferrules **32** from moving further into the loopback dust cap **20**, springs **126** that bias the connector ferrules **32** relative to the connector bodies **34** are compressed to allow the connector bodies **34** to move further into the cavity **104**. Eventually the latching features **124** on the latch arms **36** of the connector sub-assemblies **30a**, **30b** reach a position where they engage the corresponding the latching features on the body **22**. Persons skilled in optical connectivity will appreciate how this insertion of the connector **12** into the loopback dust cap **20** and engagement between the components is similar to the insertion of the connector **12** into an adapter (not shown). In alternative embodiments, the first and second cap ferrules **64a**, **64b** may be spring-biased relative to the insert **22**.

[0041] FIG. **9** also illustrates how the loopback optical fiber **66** extends between the first and second cap ferrules **64a**, **64b** in a region of the body **22** that is between the front end **100** of the body **22** and the insert **24**. The cavity **104** is still present in this region and accommodates the loopback optical fiber **66**. In the embodiment shown, the loopback optical fiber **66** extends axially from the first and second cap ferrules **64a**, **64b** toward the front end **100** of the body **22** and then makes a single bend so as to have a U-shaped configuration within the cavity **104**. The bend therefore has a diameter that approximately corresponds to the pitch between the first and second cap ferrules **64**, **64b**. Because the pitch between the first and second cap ferrules **64a**, **64b** may be relatively small (e.g., 6.25 mm for an LC-type interface and even smaller for VSFF embodiments), it may be advantageous for the loopback optical fiber **66** to be an optical fiber with low bend loss. In some embodiments, for example, the loopback optical fiber **66** may be an optical fiber that meets

(i.e., achieves or exceeds) the bend performance specifications of International Telecommunication Union standard ITU-T G.657.A2 and/or ITU-T G.657.B2, such as Corning® ClearCurve® LBL optical fiber or Corning® SMF-28® Contour optical fiber. In some embodiments, the loopback optical fiber **66** may even meet the bend performance of International Telecommunication Union standard ITU-T G.657.B3, using for example Corning® ClearCurve® ZBL optical fiber.

[0042] In use, the loopback dust cap **20** can function in the manner described at the beginning of this Detailed Description. Specifically, when an optical signal is transmitted by an optical fiber **130** terminated by one of the connector ferrules **32**, the optical signal is transmitted to the end of the loopback optical fiber **66** that is terminated by the first or second cap ferrule **36a**, **36b** (whichever is aligned and in contact with the transmitting connector ferrule **32**). The optical signal then travels through the loopback optical fiber **66** so that the loopback assembly **50** directs the optical signal into the optical fiber **130** that is terminated by the other connector ferrule **32**. At an opposite end of the optical link that includes the cable assembly **10**, the return optical signal can be compared to the transmitted optical signal to evaluate the optical link for attenuation issues.

[0043] Instead of or in addition to being used for returning an optical signal to the connector **12**, in some embodiments the loopback optical fiber **66** may be configured to scatter or otherwise emit visible light for identification purposes. For example, in some embodiments the body **22** of the loopback dust cap **20** may comprise a translucent or transparent material, such as a polycarbonate. A visual fault locator (VFL; not shown) may be used to deliver optical signals in the form of visible light to the loopback dust cap **20**, and the loopback optical fiber **66** may then emit that visible light so that the light that can be seen through the body **22** of loopback dust cap **20**. The visible light may even cause the body **22** to glow in some embodiments, thereby further assisting with identification.

[0044] While the present disclosure has been illustrated by the description of specific embodiments thereof, and while the embodiments have been described in considerable detail, it is not intended to restrict or in any way limit the scope of the appended claims to such detail. The various features discussed herein may be used alone or in any combination within and between the various embodiments. Additional advantages and modifications will readily appear to those skilled in the art. The present disclosure in its broader aspects is therefore not limited to the specific details, representative apparatus and methods and illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the scope of the present disclosure.

Claims

1. A loopback dust cap for a fiber optic cable assembly that includes a first optical fiber, a second optical fiber, and a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber, the loopback dust cap comprising: a body having a front end, a back end, and a cavity between the front end and the back end, wherein the back end of the body includes an opening into the cavity so that the cavity is configured to receive the fiber optic connector; an insert received in the cavity of the body; first and second sleeves supported by the insert and each including a respective first end facing the front end of the body and a respective second end facing the back end of the body; a first cap ferrule extending through the first end of the first sleeve, wherein the first cap ferrule includes a first end face positioned within the first sleeve; a second cap ferrule extending through the first end of the second sleeve, wherein second cap ferrule includes a second end face positioned within the second sleeve; and a loopback optical fiber having a first end terminated by the first cap ferrule and a second end terminated by the cap second ferrule, wherein the loopback optical fiber extends between the first cap ferrule and the second cap ferrule in a region of the body that is between the front end of the body and the insert; wherein the first sleeve and the second sleeve are configured

to: receive the first connector ferrule and the second connector ferrule, respectively, when the fiber optic connector is received in the cavity of the body, and align the first connector ferrule and the second connector ferrule with the first cap ferrule and the second cap ferrule, respectively.

2. The loopback dust cap of claim 1, further comprising: a first ferrule holder received over a portion of the first cap ferrule that is not within the first sleeve; and a second ferrule holder received over a portion of the second cap ferrule that is not within the second sleeve, wherein the first ferrule holder and the second ferrule holder are at least partially retained within the insert.

3. The loopback dust cap of claim 2, wherein: the insert includes a support wall, a first boss extending from the support wall toward the front end of the body, and a second boss extending from the support wall toward the front end of the body; the first boss extends over and beyond the first end of the first sleeve, and receives and retains at least a portion of the first ferrule holder; and the second boss extends over and beyond the first end of the second sleeve, and receives and retains at least a portion of the second ferrule holder.

4. The loopback dust cap of claim 3, wherein: the first boss and the second boss each comprise a respective split end portion configured to expand radially to allow insertion of the first ferrule holder and the second ferrule holder, respectively, when the insert is not received in the cavity, and the body is configured to limit or prevent the split end portions from expanding radially when the insert is received in the cavity.

5. The loopback dust cap of claim 1, wherein the insert is configured to limit or prevent the first ferrule holder and the second ferrule holder from rotating relative to the insert.

6. The loopback dust cap of claim 1, wherein the insert is coupled to the body by a snap-fit.

7. The loopback dust cap of claim 1, wherein the body includes at least one latching feature configured to engage a corresponding latching feature on the fiber optic connector when the fiber optic connector is received in the cavity.

8. The loopback dust cap of claim 1, wherein the first cap ferrule and the second cap ferrule each comprise a ceramic ferrule having a nominal diameter of 1.25 mm.

9. The loopback dust cap of claim 1, wherein the body and the insert are shaped to define an LC-type interface according to IEC 61754-20:2012 or TIA-604-10-C.

10. The loopback dust cap of claim 1, wherein the first sleeve and the second sleeve each comprise a split sleeve.

11. The loopback dust cap of claim 1, wherein the first cap ferrule and the second cap ferrule are not spring-biased relative to the insert.

12. The loopback dust cap of claim 1, wherein the loopback optical fiber meets bend performance specifications of International Telecommunication Union standard ITU-T G.657.A2 or ITU-T G.657.B2.

13. The loopback dust cap of claim 1, wherein the loopback optical fiber meets bend performance specifications of International Telecommunication Union standard ITU-T G.657.B3.

14. The loopback dust cap of claim 1, wherein the body comprises a translucent or transparent material, and wherein the loopback optical fiber is configured to emit visible light.

15. A fiber optic cable assembly, comprising: a first optical fiber and a second optical fiber; a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber; and a loopback dust cap received on the fiber optic connector, wherein the loopback dust cap includes: a body having a front end, a back end, and a cavity between the front end and the back end, wherein the back end of the body includes an opening into the cavity; an insert received in the cavity of the body; first and second sleeves supported by the insert and each including a respective first end facing the front end of the body and a respective second end facing the back end of the body; a first cap ferrule extending through the first end of the first sleeve, wherein the first cap ferrule includes a first end face positioned within the first sleeve; a second cap ferrule extending through the first end of the second sleeve, wherein second cap ferrule includes a second end face positioned within the second sleeve;

and a loopback optical fiber having a first end terminated by the first cap ferrule and a second end terminated by the cap second ferrule, wherein the loopback optical fiber extends between the first cap ferrule and the second cap ferrule in a region of the body that is between the front end of the body and the insert; wherein: the first connector ferrule extends into the first sleeve and contacts the end face of the first cap ferrule, and the second connector ferrule extends into the second sleeve and contacts the end face of the second cap ferrule.

16. The fiber optic cable assembly of claim 15, wherein the loopback dust cap further includes: a first ferrule holder received over a portion of the first cap ferrule that is not within the first sleeve; and a second ferrule holder received over a portion of the second cap ferrule that is not within the second sleeve, wherein the first ferrule holder and the second ferrule holder are at least partially retained within the insert.

17. The fiber optic cable assembly of claim 16, wherein: the insert of the loopback dust cap includes a support wall, a first boss extending from the support wall toward the front end of the body, and a second boss extending from the support wall toward the front end of the body; the first boss extends over and beyond the first end of the first sleeve, and receives and retains at least a portion of the first ferrule holder; and the second boss extends over and beyond the first end of the second sleeve, and receives and retains at least a portion of the second ferrule holder.

18. The fiber optic cable assembly of claim 17, wherein: the first boss and the second boss each comprise a respective split end portion configured to expand radially to allow insertion of the first ferrule holder and the second ferrule holder, respectively, when the insert is not received in the cavity, and the body is configured to limit or prevent the split end portions from expanding radially when the insert is received in the cavity.

19. A method of making a loopback dust cap for a fiber optic cable assembly that includes a first optical fiber, a second optical fiber, and a fiber optic connector having a first connector ferrule that terminates the first optical fiber and a second connector ferrule that terminates the second optical fiber, the method comprising: (a) assembling first and second sleeves with an insert so that the first and second sleeves are supported by the insert; (b) providing a loopback assembly that includes a loopback optical fiber having one end terminated by a first cap ferrule and another end terminated by a second cap ferrule; (c) extending the first cap ferrule through a first end of the first sleeve so that an end face of the first cap ferrule is positioned within the first sleeve; (d) extending the second cap ferrule through a first end of the second sleeve so that an end face of the second cap ferrule is positioned within the second sleeve; and (e) positioning the insert in a cavity of a body, wherein: the body includes a front end and a back end, the back end includes an opening into the cavity, and the positioning occurs after steps (a)-(d) so that the positioning results in: the respective first ends of the first and second sleeves facing the front of the body and respective second ends of the first and second sleeves facing the back end of the body, and the loopback optical fiber extending between the first cap ferrule and the second cap ferrule in a region of the body that is between the front end of the body and the insert, wherein the cavity of the body is configured to receive the fiber optic connector through the opening in the back end of the body, and wherein the first sleeve and the second sleeve are configured to: receive the first connector ferrule and the second connector ferrule, respectively, when the fiber optic connector is received in the cavity, and align the first connector ferrule and the second connector ferrule with the first cap ferrule and the second cap ferrule, respectively.
